Week 5 Assignment Run One Validation Test

Validation Type: Target-User Problem Review

Critical Hypothesis: The students who have encountered the sanitation issues in the public areas would like to take actual actions for a transparent, reliable and low-conflict cleaning service.

Main Method: Telephone Interviews

Auxiliary Method: General Questionnaires

Metric:

M-1. The Product Feasibility Index PFI of MCDA;

M-2. The Static Payback Period PP.

A-1. The Total Valid Paying Customer Rate R via Marketing Funnel;

A-2. The Matching Degree M of the Pain Points.

Threshold:

M-1. $PFI > 75.0$;

M-2. $PP < 90d$.

A-1. $R > 0.65$;

A-2. $M > 0.8$.

(The product could be viewed as "feasible" if all the main metrics are reached while at least one of the auxiliary thresholds is reached.)

Raw results: Given the data gathered via 5 telephone interviews and the 13 valid general questionnaires, we obtain:

$$PFI \approx 86.3 > 75.0$$
$$PP \approx 56.8 < 90(d)$$
$$R \approx 0.385 \not> 0.65$$
$$M \approx 0.950 > 0.8$$

Analysis: Merely an auxiliary index R fails to meet the standard, other three all perfectly reached the thresholds.

Next Decision: We could implement our preset MVP in our own campus and actively collect the post-service customers' feedback. We could also use statistical tools to reanalyze the data and polish our project.

Appendix:

M-1. $PFI = \sum_{i=1}^{n} w_i S$, in which w signifies the weight and we select $S = 100$.

M-2. $PP = \frac{I}{(p-vc)\cdot Q_{month} - FC_{month})}$, in which I signifies the initial disposable investment, p signifies the price per service, vc signifies the variable cost at a single time, Q_{month} signifies the quantity of the orders per month and FC_{month} signifies the fixed operation cost per month. We change the unit of PP from month to day in the final output.

A-1. $R = \prod_{i=1}^{n} r_i$, in which r means the ratio in every stages of the funnel.

A-2. $M = \frac{\sum_{i=1}^{m} (p_i \times c_i)}{\sum_{i=1}^{m} p_i}$, in which p signifies the referral rate and c signifies the solving rate.